Validity Is Not Difficulty: Supplementary Material

Anonymous CVPR submission

Paper ID *****

A. SceneIR Schema and Coordinate Conventions

SceneIR records explicit $T_{\text{city}\leftarrow\text{ego}}$, $T_{\text{ego}\leftarrow\text{sensor}}$, and $T_{\text{city}\leftarrow\text{actor}}$ transforms. A return $x_{s,t}$ is fused in Actor coordinates as

$$x_i = T^{-1}_{\text{city}\leftarrow\text{actor},t} T_{\text{city}\leftarrow\text{ego},t} T_{\text{ego}\leftarrow\text{sensor},t} x_{s,t}. \quad (1)$$

The physical layer stores Actor-local surfels, known/unknown masks, normals, temporal and view support, matched rays, and source timestamps. Appearance primitives are referenced only through attachment provenance; Gaussian opacity is never read as collision occupancy. Actor identity, category, extent, trajectory, lifecycle, and analytic hazard fields are immutable across compiler actions.

Four actions. KEEP preserves supported query evidence. PROJECT moves an artifact only to its matched observed termination. COMPLETE adds a canonical surfel only when temporal and view support satisfy the frozen contract. UNKNOWN is the fallback when evidence is insufficient; it never becomes free.

B. Matched-Ray Physical Certificate

Automotive LiDAR is 2.5D ray evidence, not merely an unordered 3D point set [2]. For origin o, direction v, and observed termination d, the measured free interval is

$$\mathcal{F}(o, v, d) = \{o + sv \mid 0 \le s < d - \epsilon\}. \quad (2)$$

A provenance-bearing PROJECT candidate is placed at the same ray's termination $o + dv$. It therefore cannot lie in $\mathcal{F}$. A candidate without an exact ray is marked UNKNOWN; nearest-surface projection is not substituted. This is the source of the paired zero-free-space result. It is a deterministic contract for matched rays, not a completeness proof for the whole Actor surface or a probabilistic sensor-noise guarantee.

The held-out targets are later frames excluded from both canonical fusion and query compilation. Depth, ray-termination, Chamfer, precision, and recall are computed against these target-only returns. This separation explains why zero paired free-space violation can coexist with the observed 17.19% per-Actor Chamfer tail.

C. Frozen Cohorts and Exact-Once Rules

After a quality read, scenes/logs cannot be replaced, failed cases remain in the denominator, and no threshold, radius, feature group, or checkpoint may change. Download/resource failures are reported separately from scientific failures. The fresh AV2 protocol is bound to the already frozen P6-C/P4 artifacts and is not changed by the rejected fresh nuScenes result.

D. Actor-Level Decision Intervals

For the frozen ReLU validity MLP, an input box $[\ell, u]$ is propagated through each affine layer using

$$\ell' = W^+\ell + W^-u + b, \quad u' = W^+u + W^-\ell + b, \quad (3)$$

followed by endpoint-wise ReLU. Here $W^+ = \max(W, 0)$ and $W^- = \min(W, 0)$. The resulting logit interval is mapped through the sigmoid. An Actor is *robust-select* if its lower probability is above the frozen P4 threshold, *robust-abstain* if its upper probability is below threshold, and unresolved otherwise. This is deterministic interval-bound propagation over a specified box, related to robust-OOD certification [1]; it does not assert that the box describes an unseen sensor.

We fix radii $0.05/0.10/0.20$ in nuScenes-training standard deviations. The sensor-opportunity group contains point/surfel counts, temporal/view support, range, and frame count; the physical-surface group contains surface residuals, keep/unknown fractions, completion fraction, and compiled/query ratio. Their union is exactly the 13D validity input.

At radius 0.10, 33.33% of nominal nuScenes-test selections and 91.23% of consumed-AV2 selections are robust-select. Nevertheless, 10.76% of the latter are retained target false repairs. Query-point count is the largest one-feature interval source for 83.33%/95.11% of nuScenes/AV2 Actors. Thus a shifted model can be confidently and locally stable while wrong.

Table 1. Cohort construction is metadata-only. No quality, model score, RGB appearance, or target metric is read before freezing.

Cohort	Size	Frozen rule
AV2 consumed external	30 logs	Sort all 150 Sensor-val UUIDs; take indices $0, 5, \ldots, 145$. First 20 are quantitative and final 10 qualitative.
AV2 fresh exact-once	20 logs	Remove the consumed indices; from the 120-log complement take positions $0, 6, \ldots, 114$.
nuScenes fresh exact-once	20 scenes	Exclude every P4 train/calibration/test scene; sort the remaining 106 available scenes by official index and take rounded equally spaced positions.

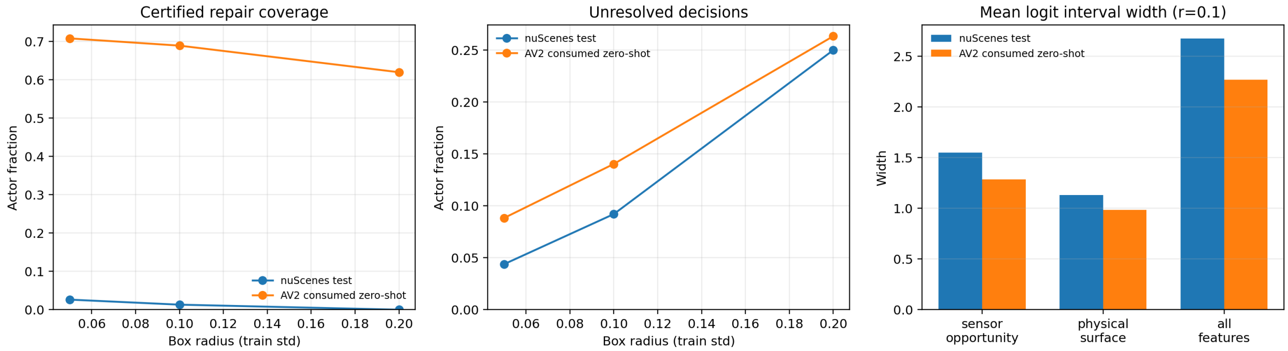

Figure 1. Frozen P7-C interval audit. Left/center: robust-select coverage and unresolved fraction across pre-registered radii. Right: sensor-opportunity boxes induce wider logit intervals than physical-surface boxes. The high AV2 certified coverage does not erase the 47 stable false repairs.

E. Failure Ledger and Claim Corrections

Table 2. Paper-facing failures are retained as boundaries rather than hidden by replacement experiments.

ID	Observation	Resulting boundary or correction
V7-F09	Aggregate geometry hides 17.19% Actor-level Chamfer worsening.	P3 is aggregate hard evidence; P4 adds repair-or-abstain and preserves all Actors.
V7-F11	AV2 validity attribution is dominated by sensor opportunity.	Zero hazard leakage does not imply domain-invariant validity.
V7-F13	Exact P4-train overlap is only two scenes/five Actors.	Reject joint fitting; audit the frozen physical/reliability interface only.
V7-F14	FP32 exceeds the analytic tolerance by 3.8×10^{-6}.	Recompute the same bound in FP64; do not loosen the tolerance or alter rows.
V7-F15	Fresh P6-C repair AUROC loses 3.50 points to P4.	Reject recovery despite higher operating-point coverage; P4 remains primary.
V7-F16	Freeze prose mislabeled the P346 held-out horizon.	Executable artifact is 2.5 s; correct documentation without rerunning.
V7-F17	47 AV2 false repairs remain robust-select at radius 0.10.	Network stability is not correctness, calibration, or a safety certificate.

F. Frozen AV2 Camera-Evidence Gallery

The following ten cases are the actor-rank-0 case from each qualitative log in the pre-registered order, not a metric or appearance ranking. Every panel shows synchronized RGB, the paired synthetic artifact probe, the four-action physical output, and sparse camera depth. The full package contains 30 panels and 30 before/after videos (three lexically first eligible Actors per log). Videos diagnose temporal behavior but are not photorealistic renders.

Figure 2. Frozen case q00, hazardous Actor 1f197615, ring-front-center; 251 visible query returns and 2,287 sparse-depth points in crop.

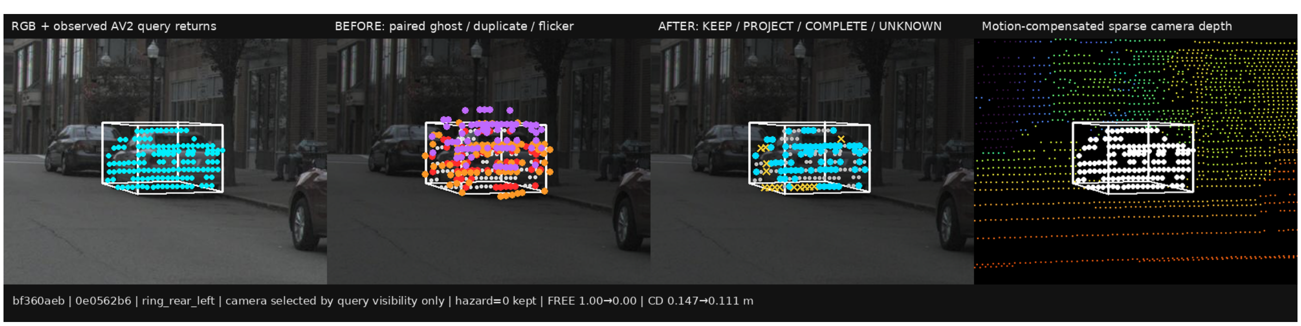

Figure 3. Frozen case q01, non-hazardous Actor 0e0562b6, ring-rear-left; 137 visible query returns and 1,770 sparse-depth points.

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Figure 4. Frozen case q02, non-hazardous Actor 08fc423e, ring-side-right; 85 visible query returns and 2,476 sparse-depth points.

Figure 5. Frozen case q03, hazardous Actor 05cccb41, ring-side-left; 2,456 visible query returns and 17,286 sparse-depth points.

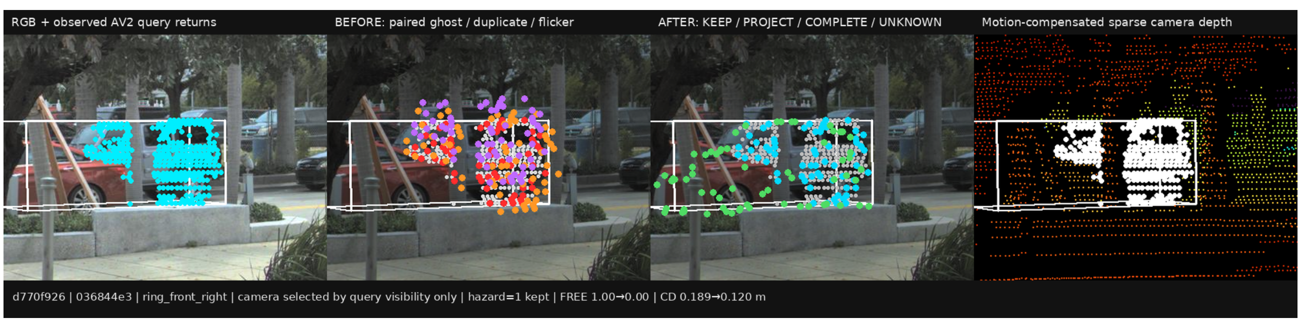

Figure 6. Frozen case q04, hazardous Actor 036844e3, ring-front-right; 292 visible query returns and 2,358 sparse-depth points.

Figure 7. Frozen case q05, hazardous far/sparse Actor 07695d24, ring-front-center; all 14 query returns remain visible.

Figure 8. Frozen case q06, non-hazardous Actor 08515e0d, ring-front-right; 79 visible query returns and 2,757 sparse-depth points.

Figure 9. Frozen case q07, non-hazardous Actor 076c9103, ring-front-center. This unfiltered case exposes Chamfer worsening and is retained.

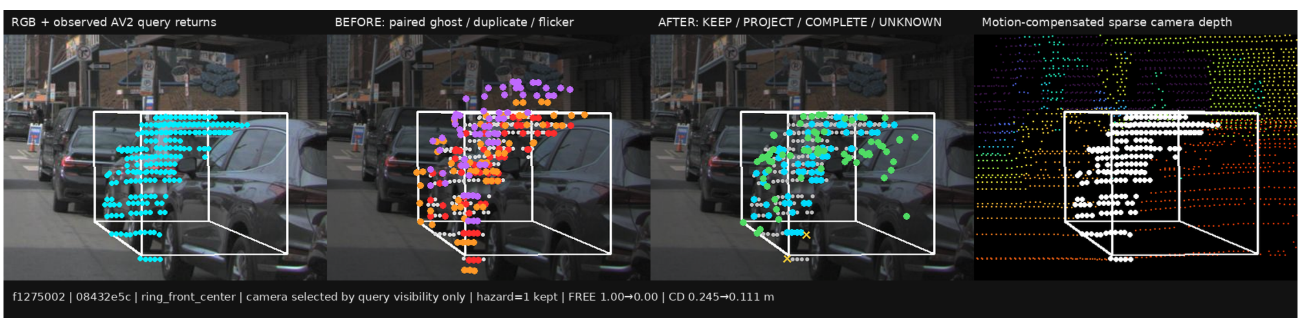

Figure 10. Frozen case q08, hazardous Actor 08432e5c, ring-front-center; 231 visible query returns and 2,101 sparse-depth points.

Figure 11. Frozen case q09, non-hazardous Actor 6d55ff95, ring-front-center; 38 visible query returns and 1,630 sparse-depth points.

G. Resource and Reproducibility Boundary

All V7 scientific runs use one RTX 3090. The largest paper-facing GPU allocation is below 1.5 GiB, while P7-C uses 9.2 MiB and 1.37 s. Download and preprocessing are sequential on the low-memory host. Artifacts are identified by task/run paths and immutable cohort/config files; the protocol intentionally does not add hashes, checksums, fingerprints, repeated smoke matrices, or post-hoc quality gates.

References

[1] Julian Bitterwolf, Alexander Meinke, and Matthias Hein. Certifiably adversarially robust detection of out-of-distribution data. In *Advances in Neural Information Processing Systems*, 2020. 1

[2] Peiyun Hu, Jason Ziglar, David Held, and Deva Ramanan. What you see is what you get: Exploiting visibility for 3d object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11001–11009, 2020. 1